

DSA **Binary Search** Exercise: part-1

Q1. Normal Binary Search:

Input: nums = [-1,0,3,5,9,12], target = 9

Output: 4

Explanation: 9 exists in nums and its index is 4

Q2. Binary Search on ascending order:

Input: nums = [4,5,6,7,0,1,2], target = 0

Output: 4

Explanation: after search 0 exists in nums and its index is 4

Q3. First Occurrence:

Input: nums = [5,7,7,8,8,10], target = 8

Output: 3

Explanation: First occurrence of 8 is at index 3

Q4. First & Last Occurrence:

Input: arr[] = [1, 3, 5, 5, 5, 5, 67, 123, 125], x = 5

Output: [2, 5]

Explanation: First occurrence of 5 is at index 2 and last occurrence of 5 is at index 5

Q5. Find Peak in a mountain array:

Input: arr = [1, 2, 4, 5, 7, 8, 3]

Output: true

Explanation: arr[5] = 8 is a peak element because arr[4] < arr[5] > arr[6].

Q6. Floor in a sorted array:

Input: arr[] = [1, 2, 8, 10, 10, 12, 19], x = 11

Output: 4

Explanation: Largest Number less than or equal to 11 is 10, whose indices are 3 and 4. The index of last occurrence is 4.

Q7. Maximum count of positive integer and negative integer:

Input: nums = [-3,-2,-1,0,0,1,2]

Output: 3

Explanation: There are 2 positive integers and 3 negative integers. The maximum count among them is 3.

Q8. Square Root:

Input: n = 11

Output: 3

Explanation: Since, 11 is not a perfect square, floor of square root of 11 is 3.

Will Continue...